

A12749W1

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics and Economics
Honour School of Economics and Management
Honour School of History and Economics
Honour School of Engineering, Economics and Management (Part II)
Honour School of Materials, Economics and Management (Part I)

MICROECONOMICS

OUTLINE ANSWERS TO PROBLEMS

TRINITY TERM 2017

Friday 26th May 2017, 14:30 - 17:30

Please start the answer to each question on a separate page.

There are 10 questions in this paper.

*Answer **all** questions in Part A and **two** long questions from Part B.*

Part A attracts 1/3 of the marks in total; each question in Part B attracts 1/3 of the marks.

In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets;

in Part B, the weight assigned to each part of the multipart question is indicated in square brackets.

Candidates may use their own calculators.

Do not turn over until told that you may do so.

Part A

In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets

1. Consider an economy which produces output X using labour input L according to the production function $X = 2L^{1/2}$, and which contains one consumer with the utility function $u = X - \frac{1}{2}L^2$. [30%]

- (a) If the consumer controls production directly, what quantity of labour input will the consumer choose?

$$u = 2L^{1/2} - \frac{1}{2}L^2, \text{ so } u' = L^{-1/2} - L, \text{ which is 0 when } L = 1.$$

- (b) Define a general competitive equilibrium.

bookwork: price vector and allocation s.t.

- firms max profits,
- profits distributed to households,
- households max utility subject to budget constraint,
- all markets clear.

- (c) Suppose now that the price of X is p and the price of L is w .

Production is chosen by a firm with the production function given above to maximize profits. Derive the firm's labour demand function and its level of profits as functions of p and w .

The household has the utility function given above, which it maximizes subject to its budget constraint. Derive the household's labour supply function.

What is the equilibrium price-wage ratio of this economy, and what is the equilibrium level of employment and output?

Explain your result in terms of the first theorem of welfare economics.

Outline briefly how your answer would change if there were a tax on wage income in this economy.

$$\pi = p2L^{1/2} - wL, \text{ so } \pi' = pL^{-1/2} - w, \text{ which is 0 when } L = (p/w)^2, \\ \text{giving } \pi^* = p^2/w.$$

$$u = (wL + \pi^*)/p - \frac{1}{2}L^2, \text{ so } u' = w/p - L, \text{ which is 0 when } L = w/p.$$

$$\text{Equilibrium when labour supply = demand: } (p/w)^2 = w/p, \text{ i.e. } p/w = 1.$$

$$\text{So in equilibrium, } L = 1 \text{ and } X = (wL + \pi^*)/p = 2.$$

Competitive equilibrium supports the (Pareto) efficient outcome.

A tax on wage income would place a wedge between the consumer's MRS and the producer's MRT.

2. Two firms sell identical products and are the only firms in the industry. They have the same constant and positive marginal cost, no fixed costs, and they set prices non-cooperatively.

Explain why, in theory, a merger between the duopolists creating a profit-maximizing monopolist could increase social welfare.

Could such a merger also benefit consumers?

(A well-annotated diagram could form the basis of your answer.)

[15%]

With identical constant marginal costs, no fixed costs, and identical products, the Bertrand-Nash equilibrium applies: both prices will equal marginal cost. A merger that gives the firms monopoly power will raise price to the monopoly level if costs are unchanged -- that would reduce both social welfare and consumer surplus.

However, if the merger cuts the marginal cost sufficiently then the increase in profits can offset the reduction in consumer surplus.

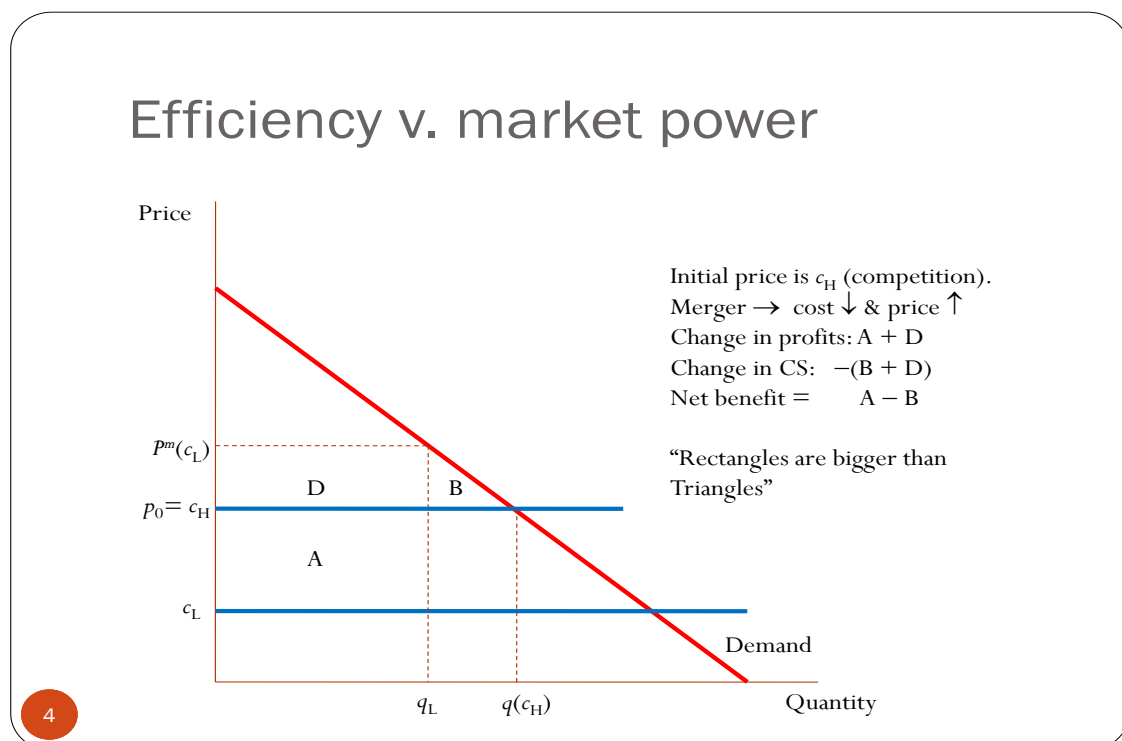
This is the Williamson trade-off;

the maxim 'rectangles are bigger than triangles' is sometimes applied.

The best way to show this is to use an accurate monopoly diagram.

If the cost reduction is sufficiently large the monopoly price will be below the old marginal cost, and consumers would then benefit from a lower price.

(In IO this is known as a "drastic innovation", but candidates do not need to know the name.)



3. Janet's utility as a function of wealth is $u(w) = 2w^{1/2}$. [25%]

- (a) Show that her coefficient of absolute risk aversion is decreasing in initial wealth, but her coefficient of relative risk aversion is constant.

The coefficient of absolute risk aversion (at w) is $A(w) = -\frac{u''(w)}{u'(w)}$.
 $u'(w) = w^{-1/2}$, $u''(w) = -\frac{1}{2}w^{-3/2}$, so $A(w) = \frac{1}{2}w^{-1}$, which is decreasing in w .

The coefficient of relative risk aversion (at w) is $R(w) = -\frac{w u''(w)}{u'(w)} = wA(w)$.
 So here, $R(w) = \frac{1}{2}$.

- (b) She has initial wealth $w = 104$ and is facing the lottery $(\frac{1}{2}, -40; \frac{1}{2}, +40)$.

What is her certainty equivalent of this lottery, and what is the associated risk premium?

The expected utility of the lottery is

$$\bar{u} = \frac{1}{2}u(104 - 40) + \frac{1}{2}u(104 + 40) = \frac{1}{2} \times 2(64)^{1/2} + \frac{1}{2} \times 2(144)^{1/2} = 8 + 12 = 20.$$

The certainty equivalent satisfies $u(\text{ce}) = \bar{u}$, so $2\text{ce}^{1/2} = 20$, $\text{ce} = 100$.

The risk premium is the difference between the expected value of the outcomes and the certainty equivalent: $\text{rp} = \bar{y} - \text{ce} = 104 - 100 = 4$.

- (c) Sam has the same utility function and initial wealth, but is instead facing the lottery $(\frac{1}{2}, -20; \frac{1}{2}, +20)$.

What, *approximately*, is the ratio of Sam's risk premium to that of Janet from the previous part, and why?

(You should not recalculate the certainty equivalent for Sam's lottery.)

The risk premium is approx. $\frac{1}{2}A(w)\sigma^2$, where σ^2 is the variance of the gamble.
 The standard deviation, σ , of Sam's lottery is 1/2 that of Janet's,
 so the rp will be approximately 1/4 of Janet's.

>>> ADDED LATER

This is nothing to do with the power of 1/2 in the utility function,
 nor the fact that the coeff. of RRA is the same for both S & J.

4. There are two types of worker: H -types with high productivity θ_H , and L -types with low productivity θ_L ; each worker knows their own type. Many risk-neutral firms compete for the services of the workers, but they cannot observe a worker's type. [30%]

- (a) Why would some of the workers like to be able to signal their type to the firms?
What condition(s) would make a signal credible?

H -types would like to signal and get a wage $w = \theta_H$ rather than average productivity, if the cost of the signal is less than the wage difference.
To be credible, signal must be too expensive for L -types to copy.

Suppose that $\theta_H = 240$, $\theta_L = 180$, and that there are half as many H -types as L -types; also, the workers have no outside option.

- (b) When no signal is available, what is the market equilibrium outcome?

Expected productivity: $\frac{1}{3}\theta_H + \frac{2}{3}\theta_L = \frac{1}{3} \times 240 + \frac{2}{3} \times 180 = 80 + 120 = 200$.

All workers are employed, at a wage of 200.

Suppose the workers can acquire an education, observable by the firms – this would cost only 45 for an H -type, but 55 for an L -type.

- (c) Consider an outcome in which H -types signal and L -types don't, the firms paying appropriate wages.

Is this outcome a market equilibrium? If not, what *is* the equilibrium outcome?

Is the equilibrium efficient?

If the firm pays $w_H = 240$ to a worker that signals, and $w_L = 180$ to one that doesn't, then L -types have a profitable deviation, since $240 - 55 = 185 > 180$.

>>> ADDED LATER

NOT: H -types have a profitable deviation to pooling and receiving 200!!!

The only equilibrium has the workers pooling by not signalling and then receiving a wage of 200.

(H -types wouldn't deviate -- at best they would net $240 - 45 = 195 < 200$.)

In this case, the equilibrium is efficient and is the same as when signalling is unavailable.

>>> ADDED LATER

There are also two semi-separating equilibria:

- L 's don't signal, H 's don't signal w.p. α ;
expected productivity = $(\frac{2}{3}\theta_L + \alpha\frac{1}{3}\theta_H)/(\frac{2}{3} + \alpha\frac{1}{3})$;
 H 's willing to mix when this = $\theta_H - 45$, so $\alpha = 2/3$.

- H 's signal, L 's signal w.p. β ;
expected productivity = $(\beta\frac{2}{3}\theta_L + \frac{1}{3}\theta_H)/(\beta\frac{2}{3} + \frac{1}{3})$;
 L 's willing to mix when this = $\theta_L + 55$, so $\beta = 1/22$.

Part B

10. A risk-neutral farm owner (the principal) hires a risk-averse farm worker (the agent) to manage this year's crops. The harvest will be either small, worth only 10 to the farm owner, or large, worth 50. The size of the harvest depends on which of three effort levels that the worker exerts, as summarised in the table.

	small (10)	large (50)
Pr e_H	1/5	4/5
Pr e_M	4/5	1/5
Pr e_L	1	0

The worker's utility from receiving a wage w and exerting effort e is $\sqrt{w} - e$, where $e = 3$ if effort is e_H , 2 if effort is e_M , and 1 if effort is e_L . The worker has a reservation utility of 0.

- (a) When both effort and the size of the harvest are observable, what does the owner consider when deciding which contract to offer the worker?

What contract will the owner choose? [20%]

The owner will offer a wage $w|e = e^2$, i.e. 1, 4 or 9;

$E[\pi|e_L] = 10$, $E[\pi|e_M] = 18$, $E[\pi|e_H] = 42$.

Optimal contract is $e = e_H, w = 9$.

Now assume that effort is unobservable. The wage in the contract between the two parties will depend only on the size of the harvest: w_1 if it is a small harvest, w_2 if it is large.

- (b) Write v_i for the value to the worker of a wage w_i , i.e. $v_i = \sqrt{w_i}$.

If the worker would prefer to exert effort e_M rather than e_H , what is the largest value that $v_2 - v_1$ can be?

If the worker would prefer to exert effort e_M rather than e_L , what is the smallest value that $v_2 - v_1$ can be?

Hence, or otherwise, show that there is no contract under which it is incentive compatible for the worker to choose e_M . [30%]

For $e_M \succsim e_H$: $\frac{4}{5}v_1 + \frac{1}{5}v_2 - 2 \geq \frac{1}{5}v_1 + \frac{4}{5}v_2 - 3$, so $v_2 - v_1 \leq 5/3$.

For $e_M \succsim e_L$: $\frac{4}{5}v_1 + \frac{1}{5}v_2 - 2 \geq v_1 - 1$, so $v_2 - v_1 \geq 5$.

Clearly, there are no values of (v_1, v_2) that satisfy both inequalities.

- (c) Explain briefly, and in general, (i) how the owner would implement low effort, e_L ,
(ii) how he would implement high effort, e_H .

What is the optimal contract in this case? [35%]

To implement low effort:

offer a fixed wage that just satisfies the Participation Constraint.

Here, the contract would be (1,1) with the expected net profit = $10 - 1 = 9$.

To implement high effort:

offer a variable wage that just satisfies the Participation Constraint,
and also just satisfies the Incentive Constraint.

So, $\frac{1}{5}v_1 + \frac{4}{5}v_2 - 3 \geq 0$, and $\frac{1}{5}v_1 + \frac{4}{5}v_2 - 3 \geq v_1 - 1$.

This leads to $v_1 = 1$ and $v_2 = 7/2$, and so $w_1^* = 1$ and $w_2^* = 49/4$.

$E[w|e_H] = 10$, and the expected net profit = $42 - 10 = 32$.

Clearly, it is better to induce high effort with the contract $(1, 12\frac{1}{4})$.

>>> ADDED LATER

Ideally an answer would note the following, implied by the previous part.

We have two contracts:

the fixed-wage one in which the agent prefers e_L to e_H ,

and the other one in which the agent prefers e_H to e_L ;

we don't need to check incentive compatibility with respect to e_M ,

since if $e_M \succsim e_L$ in the fixed-wage contract or $e_M \succsim e_H$ in the other contract
then it's preferred to both (by transitivity) and that's not possible.

Alternatively (but less elegant),

calculate that $v_2 - v_1 > 5/3$ in the contract that the owner prefers,

so $e_H \succ e_M$.

- (d) Why does the owner suffer from effort being unobservable, and by how much? [15%]

In order to implement e_H , the expected wage is higher when effort is not
observable; this is because a variable wage that assures the agent of his
reservation utility is greater than a fixed wage that does the same.

The agency cost is the risk premium associated with the lottery for an agent
who exerts effort e_H :

$E[w|e_H] = 10$, whereas with observable effort the fixed wage is 9;

so the agency cost is 1.